EC2 Bootstrap Script (Ubuntu 24.04 LTS)

Proposed Solution

An idempotent Bash script (bootstrap.sh), run at every boot by a systemd service. It installs the latest OpenJDK LTS if Java is missing, enables automatic security updates, and installs the Datadog Agent with the API key read from SSM Parameter Store at runtime.

Assumptions

- If any Java is already installed, the script leaves it alone.
- The API key is an SSM SecureString at /datadog/api_key, encrypted with a customer-managed KMS key in the same region as the instances.
- Instances have outbound internet access.
- Security updates do not reboot the instance automatically.

Why Not Other Options

- **cloud-init per-boot scripts:** no retry, and logs are harder to find.
- **default-jdk:** installs OpenJDK 21, while Ubuntu 24.04 already offers OpenJDK 25.

Running It at Every Startup

cloud-init user data installs the script and this unit on first launch and enables it. systemd then runs it on every boot.

[Unit]

Description=EC2 bootstrap
Wants=network-online.target
After=network-online.target snapd.seeded.service
[Service]
Type=oneshot
ExecStart=/usr/local/sbin/bootstrap.sh
Restart=on-failure
RestartSec=2min
[Install]
WantedBy=multi-user.target

I chose systemd because it waits for the network, retries on failure, and logs to journald (journalctl -u bootstrap).

IAM Instance Role Permissions

See iam-policy.json. The region, account ID and KMS key ID are placeholders.

- **ssm:GetParameter** on the API key parameter only.
- **kms:Decrypt** on one key, only through SSM and only for this parameter.
- **cloudwatch:PutMetricData** in the Bootstrap namespace only.

The script reads the key with aws ssm get-parameter --with-decryption. SSM uses KMS to decrypt it with the instance role. No secret is stored in the script or the repo.

IMDSv2 needs no IAM permission. The script and the AWS CLI both use IMDSv2 tokens.

Detecting Failures Across 50 Instances

1. Every run sends a BootstrapFailed metric (0 or 1) to CloudWatch. An alarm alerts the team if any run fails.
2. A Datadog monitor alerts if fewer than 50 hosts are reporting, which catches instances where the script never ran.
3. journalctl -u bootstrap on the instance, through SSM Run Command, shows the cause.